

Micro Syllabus

Subject: Instrumentation System II/II

Marks Allotment	Theory	Practical	Total
Sessional	40		40
Final	60		60
Total	100	25	125

Time Allotment	Lecture	Tutorial	Practical
Weekly Hours			
Depth Codes	C – Circuit	D – Definition	Dm – Demonstration
Dr – Derivation	Dw – Drawing	E – Explanation	I – Illustration
N – Numerical	P – Proof	Pr – Programming	S – State

Unit	Topic/ Sub topic	Depth Code	Description of Depth	Actual Plan		Plan for this semester		Week
				L	T	L	T	
1	Introduction			2				1
	1.1. Analog and digital instrument: Definition, block diagram, characteristics	D, E, I	Definition of Instrumentation, Analog Instrument (Characteristics, Simple Block Diagram) Digital Instrument (Characteristics, Simple Block Diagram), Comparison, between Analog and digital instrument)	0.5				
	1.2. Microprocessor-based systems: Open vs closed loop, benefits, features and applications in instrumentation design	D, E, I	Features of MBI, Function, Benefits, Disadvantage, Application of MBI System, Open Loop: Block diagram, Explain with any example, Advantages and disadvantages Closed Loop :Block diagram, Explain with any example, Advantages and disadvantages	1				
	1.3. Microcomputer on instrumentation design	D, E, I	Block Diagram ,explanation, factor considerations while selections a processor	0.5				

2	Theory of Measurement			4	2			2,3
	2.1 Static performance parameters: Accuracy, precision, sensitivity, resolution and linearity	D, E	Definition of Measurement, Introduction to Analog and Digital measurement, Static Performance (All Definition, Types (If any) brief explanation.	1				
	2.2 Dynamic performance parameters: Response time, frequency response and bandwidth	D, E,	All, Definition, brief explanation.	0.5				
	2.3 Error in measurement	D, E, N	General Error: (Gross Error, Systematic[Types] Error, Random Error) Brief Introduction Error based on statistic (Absolute error, Relative Error, Relative Accuracy) , Definition , formula , Solved Numerical		1			
	2.4 Statistical analysis of error in measurement	D, E, N	Statistical Analysis : (Arithmetic Mean, Deviation, Standard Deviation, Variance and Geometric Mean) Definition, Formula Solved Numerical		1			

2.5	Measurement of resistance (Low, medium and high), Measurement of Medium Resistance (Ammeter – Voltmeter, Wheatstone bridge [DC Bridge])	D, E, Dr, C, N	Introduction, Differentiate among Low, Medium and High Resistance. Measurement of medium resistance : Definition, Circuit diagram, derivation	1.5					
2.6	AC bridge(Maxwell's bridge, Schering bridge)	D, E, C, Dr, N	Definition, Circuit diagram, derivation Circuit diagram	1					

3	Transducer			8					3,4
3.1	Introduction to transducer, workflow of a transducer in typical system, transducer classification	D, E, Dr, I, C, N	Definition, Workflow, example with figure, factor consider while selecting transducer, Classification based on functional classification, output signal, power requirement, physical quantity, Resistive Transducer (PoT , Loading PoT Derivation) Numerical, Capacitive Transducer (Explain all three Principle), Numerical, Piezoelectric effect(derivation, Numerical)	2	1				
3.2	Introduction to sensor and its working principle (Resistive, capacitive and piezoelectric), generation of sensor, classification of sensor (Analog sensor, digital sensor)	D, I	Definition, Workflow diagram and explanation, Based on power requirement, Based on Measured physical quantity, Based on Principle of Operation, Based on Operation (Simple description and example), Based on Output type, Analog vs Digital Sensor, Generation of Sensor (From 1st to 5th Generation) Definition, characteristics and demerit	2					
3.3	Types of sensors (Electrical sensor, chemical sensor, biological sensor, acoustic sensor, optical sensor and other motion sensor), characteristic of sensors	D, I	All types of sensors [Definition simple working process, application] Electric sensor, Chemical sensor (e.g. any such as breathalyzer) Biological sensor, Optical Sensor, Acoustic sensor, Motion Sensor (e.g. such as Ultrasonic sensor, PIR sensor, Video Motion Sensor, Vibration Sensor), Characteristic (Static and Dynamic)	2					
3.4	Introduction to actuator, classification of actuators (Hydraulic, pneumatic, electric and mechanical), characteristic of actuator	D, E	Definition, Work flow diagram and explanation, working principle of all actuators and their application, comparison of all transducers with respect to their features, Briefly explain characteristics of actuators.	1					

4	Interfacing of Instrumentation System			10	4				5,6,7, 8,9
4.1	Microprocessor and microcontroller and their selection criteria, and applications	D, E	Microcontroller basic definition, simplified block diagram, difference between microprocessor and microcontroller. Common Microcontroller family, features, Commonly used microcontroller (brief introduction of STM-32, Arduino Atmega 328, ESP-32, Raspberry Pi, Qualcomm QRB), Selection Criteria according to features and application).	2					

4.2	The PPI 8255 and interfacing of peripherals (LED, 7 segment, dip switch, 8 bit ADC, 8/10-bit DAC using mode 0 and mode1) with 8085 microprocessor	D, E, Pr, I, N,C	Review of Handshaking, 8255 control word, BSR word, Mode 0, mode 1 (Theory and Problems) example should include LED, dip switch, 7 segment. ADC/DAC: Basic definition, Concept of status check and interrupt check, Interfacing of ADC status and ADC 0801 (Interrupt check), Interfacing of 1408 DAC.	3	2				
4.3	Microcontrollers (Atmega328, STM32): Architecture, pin configuration, and their application	D, I, ,C, Dw	Definition of microcontrollers Atmega 328 and STM32, ARM vs ARM-Cortex-M architecture based on feature. Atmega: Simplified Block-diagram Internal architecture diagram, features, pin configuration, Pin Function, Atmega328 : (Arduino Pin) Simple block diagram, Pin configuration, Pin function, Timers and Interrupt, Characteristic of 328 STM32: STM32(Cortex-m) Microcontroller, ARM cortex family, Cortex-M4 single core processor, General Architecture, Block Diagram, Pin configuration, Example of any system using STM32 Board, (such as Smart Home), Importance of ARM Cortex based microprocessor , Choosing right microprocessor of Atmega 328 and STM 32 Based.	3					
4.4	Sensor/Actuator interfacing with Atmega328P (Arduino): Analog and digital sensors, implementation of communication protocols, interrupt based interfacing	D, I, C, Pr, E	Interfacing with Atmega 328P, Minimum interfacing circuit, Prototyping breadboard connection, e.g. Hello world , Commonly used libraries in Arduino, Sensor Interfacing: Digital read concept (Switch and Led Interfacing), ADC in Atmega328 (simplified diagram), Analog Sensor, TMP36 Temperature sensor, Interfacing, LDR sensor Interfacing, Actuator Interfacing: Concept of PWM: Analog output using PWM, DC Motor using transistor (using PWM to control speed), Relay Switch, Implementation of Communication Protocol: E.g.: Interfacing TC74 Temperature Sensor using I2C, Accelerometer interfacing using SPI. Interrupt: Interrupt Concept, Polling vs Interrupt, External Hardware Interrupts, Timer Interrupt (simple timer interrupt blink test)	2	2				
5	Connectivity Technology in Instrumentation System			5	1				9,10
5.1	Introduction to wired and wireless communication system	D, E	Wired: Twisted, Coaxial and Optical fiber (Definition, Simple Characteristics, Advantage and disadvantages.) Wireless: Radio wave, Microwave, Infra-red (Definition, Simple Characteristics, Advantage and disadvantages.)	1.5					
5.2	Wired connectivity: UART, I2C, SPI, CAN	D, E, C	Synchronous transmission and Asynchronous transmission, UART (Definition, ,feature, working principle, Frame ,advantages), I2C(Definition, features, Working principle, Example ,advantages), SPI (Intro Communication codes, Hardware and communication design, Hardware configuration), Comparison UART, I2C and SPI, CAN (Brief Introduction)	1	1				

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7	Software for Instrumentation Application			6				12,13
	7.1 Introduction to software engineering	D, E	Definition, Importance of software engineering	0.5				
	7.2 Types of software	D, I, E	System software and Application software (Brief explanation with example), Hierarchy of different software language	0.5				
	7.3 Software development life cycle (SDLC), software process models	D, I, E, Dw	Engineering, Definition, Explain all Steps involving SDLC in detail. Waterfall model, Prototype Model, Incremental Model, Agile Method (Introduction, steps with relevant figure, benefits and their limitation)	3				
	7.4 Software reliability vs hardware reliability	D, E, I	Definition, Factor influencing Software reliability, fault tolerance	1				
	7.5 Software bugs, software testing, different levels of testing	D, E	Software bugs, software testing, different levels of testing Definition, Brief explanation of white Box, Black box (Functional/ Non Functional), Gray box testing. Different Level of testing: Brief Introduction, Different Level (Unit, Integration, System, Acceptance Testing), Brief Explanation of Coupling and Cohesion, Modularity.	1				
8	Electric Equipment			6				13, 14
	8.1 Wattmeter: Types and working principle	D, E, C, Dr	Introduction, Brief Explanation of types, Working principle : Electrodynamometer (Circuit Diagram, Components, Derivation)	2				
	8.2 Energy meter: Types and working principle	D, E, C, Dr	Introduction, Brief Explanation of types, Working principle : Induction Type Motor (Circuit Diagram, Components, Derivation)	2				
	8.3 Frequency meter: Types and working principle	D, E, C, Dr	Introduction, Brief Explanation of types, Working principle : Ferro dynamic type frequency meter (Figure, Phasor Diagram, Explanation)	2				
				3				14
9.	Recent Trends							
	9.1 Introduction to IOT: Simple architecture, characteristics, advantages	D, E, Dw	Definition, Architecture diagram and Illustration, Brief Explanation of Characteristic of IoT System, its advantage, Application area of IoT	1				
	9.2 Smart sensors	D, I	Structural block diagram of smart sensor, Brief explanation.	0.5				
	9.3 Important of cloud computing in instrumentation system	D, I, E	Cloud Definition, Importance (such as Management, Accessibility Scalability and so on)	0.5				
	9.4 cloud based data acquisition system	D, I, E	Explain the importance of Data acquisition System in Cloud, simple deployment figure	0.5				
	9.5 Instrumentation in industry 4.0	D, I, E	Industry revolution, Definition of Industry 4.0, Features of 4.0	0.5				

			6				
10	Application of Modern Instrumentation System		1				15
	10.1 Instrumentation for power station including all electrical and non-electrical parameters		1				
	10.2 Instrumentation for wire and cable manufacturing and bottling plant		1				
	10.3 Instrumentations for a beverage manufacturing and bottling plant		1				
	10.4 Instrumentations required for a biomedical application such as a medical clinic or hospital		1				
	10.5 Instrumentation system design using a processor (Microprocessor, microcontroller or others)		1				
	10.6 Other industries can be selected with the consent of the subject teacher or needs		1				

Reference Book:

1. Goankar, R. S., (2000). Microprocessor Architecture, Programming and Application with 8085. Prentice Hall
2. Hall, D. V., (1999). Microprocessor and Interfacing, Programming and Hardware. Tata McGraw Hill
3. Fowler, K. R., (1996). Electronic Instrument Design: Architecting for the Life Cycle. Oxford University Press, Inc.
4. Gupta, J. B., (2008). A Course in Electrical and Electronics Measurement and Instrumentation, Kataria and Sons.
5. Misra, S., Mukherjee, A., & Roy, A. (2021). *Introduction to IoT*. Cambridge University Press
6. Lutenberg, A., Gomez, P., & Pernia, E. (2022). *A beginner's guide to designing embedded system applications on Arm Cortex-M microcontrollers*. Arm Education Media.
7. Blum, J. (2020). *Exploring Arduino* (2nd ed.). John Wiley & Sons, Inc.
8. DE Silva C. W., Sensors and Actuators: Control System Instrumentation. CRC Press Taylor and French Group Boca Raton London New York.